Tianyuan (Tiana) Wang

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Summary

Ph.D. candidate in Applied Economics (STEM) with data scient training and practices and 3 years of analytical experience at the International Monetary Fund (IMF). Use **causal inference** to measure **incremental and dynamic** effects and understand **spatial** disparities and differences **over time** in **market responses and pricing**. Combine **machine learning**, **NLP**, and **scalable data pipelines** to turn **structured data**, including **NielsenIQ/Circana scanner data**, and **unstructured text data** into actionable insights and present findings to high-profile clients.

Technical Skills

Tools: Python, SQL, R, Stata, ArcGIS, Shiny, Plotly, Tableau, Power BI, AWS S3 & EC2, IBM Watson Studio.
Methods: Causal inference, DiD, RDD, bunching, local projections, time series, panel and spatial analysis, ML/NLP.
ML/NLP: Deep learning (CNNs, RNNs), tree-base supervised learning, clustering, topic modeling, sentiment analysis.

Professional Experience

International Monetary Fund | Research Analyst 2020–2023
African Department, Outstanding Performance Award 2022 *Washington, DC*

- Extract **50k+ historical lending records since 1952** from published PDF reports; rebuilt an analytical workflow for country credit status and lending program monitoring by ETL automation, **boosting ∼95% efficiency (4+hrs to ∼10mins)**; delivered monitoring dashboards for the department-wide usage. [Power BI, SQL, Python]
- Replicated financial crisis prediction models from IMF Working Paper WP/04/52 and extended them with a **sentiment index** derived from Financial Times crisis coverage; **automated ETL workflows** and backtested competing models (**logistic regression, XGBoost and OneClassSVM**) on monthly data, **benchmarking predictive performance** using confusion matrices and F1/F2 scores. [Python, Excel, Stata, Project overview].
- Developed and regularly maintained an interactive surveillance dashboards for economic diagnosis using WEO, Haver and World Bank data; drafted story-telling panel charts in IMF Article IV Staff Reports. [Excel, R, Flexdashboard]

Global Risk Intelligence | Lead Data Scientist 2019–2020
Boutique risk advisory firm *Washington, DC*

- Led a team of 3 to devise and execute a **ML-based** project analyzing supply chain risk; pitched findings to director-level clients; built dashboard **adopted by 60+ employees** at the client's risk division. [Jira, Tableau]
- Developed hypothesis-based analysis by applying **logistic/multivariate regression, decision-tree/random-forest classification and K-means/SVM clustering**; wrote structured playbooks for reproducibility and model validation. [R, Python, AWS S3 & EC2]

Selected Projects

H-2A Minimum Wages and Market Adjustment Across the Food Supply Chain Job market paper
With Michael Adjemian (dissertation advisor), presented at AAEA 2026, see manuscript

- Investigate wage-policy **dynamic market adjustment** on food prices using **LP-DiD**, distinguish **responses timing** around announcement and implementation from county seasonality and common regional shocks.
- Built county-by-category monthly price panels from an **8TB+ scanner-data** collection spanning **3,100+ U.S. counties** with R and high-performance computing cluster. [R, Linux/SLURM]

Twitter Analysis of Hong Kong 2019 Anti-extradition Bill Movement | Social Media Text Analysis
Sentiment, topic modeling and co-occurrence networks, published at GitHub

- Mining **4M+** raw tweets over 6 months through the **Twitter API** in R, converting to large corpus and document-feature matrices to examine how public discussion evolved around policy events. [R, APIs]
- Combined keyness testing (**Quanteda in R**) with sentiment consideration (**VADER in Python**), performed **Structured Topic Modeling (STM) and co-occurrence networks** to compare themes and attitudes over time; distinguished descriptive patterns from causal effects. [R, Python]

Education

University of Georgia 2023–2027 (expected)
Ph.D. in Agricultural and Applied Economics; concurrent M.S. in Statistics.

Johns Hopkins University 2018–2020
M.S. in Government Analytics (Data Analytics and Policy)

Hong Kong Baptist University 2014–2018
B.B.A. in Applied Economics; minor in Finance.